

```
#include <stdio.h>
```

```
#include <math.h>
```

```
int x[50],n,count=0;
```

```
int place(int k,int i)
```

```
{  
    int j;  
    for(j=1;j<=k-1;j++)  
    {  
        if((x[j]==i) || (abs(x[j]-i)==abs(j-k)))  
        {  
            return 0;  
        }  
    }  
    return 1;  
}
```

```
void sol()
```

```
{  
    int i,j;  
    printf("\n\nSolution #%%d:\n",count+1);  
    for(i=1;i<=n;i++)  
    {  
        for(j=1;j<=n;j++)  
        {  
            if(x[i]==j)  
            {  
                printf("Q%%d\t",i);  
            }  
            else
```

```

        {
            printf("*\t");
        }
    }
    printf("\n");
}
}

```

```

void nqueen(int k)

```

```

{
    int i;
    for(i=1;i<=n;i++)
    {
        if(place(k,i))
        {
            x[k]=i;
            if(k==n)
            {
                sol(n);
                count++;
            }
            else
            {
                nqueen(k+1);
            }
        }
    }
}
}

```

```

int main()

```

```

{

```

```
int i;  
printf("Enter the No. of Queens: ");  
scanf("%d",&n);  
nqueen(1);  
printf("The Solution is = %d",count);  
return 0;  
}
```